

Brutal Series B01 - Heat | Comparing Quantities | Electric Current | Simple Equations (~2x)

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. At dawn a laboratory thermometer reads -8°C . By noon it had risen 23°C , and by night it fell 9°C from the noon reading. What does it read at night? [\[Heat\]](#)
 - (A) 24°C
 - (B) 14°C
 - (C) -24°C
 - (D) 15°C
 - (E) 6°C
 - (F) 40°C

2. A bottle of acid costs Rs 200. The price first rises 20%, then the new price falls 20%. What is the final price? [\[Comparing Quantities\]](#)
 - (A) Rs 200
 - (B) Rs 188
 - (C) Rs 196
 - (D) Rs 160
 - (E) Rs 144
 - (F) Rs 192

3. An electromagnet picks up 6 pins when its wire has 20 turns. If the number of pins lifted is proportional to the turns, about how many pins would 50 turns lift (all else the same)? [\[Electric Current & its Effects\]](#)
 - (A) 60 pins
 - (B) 12 pins
 - (C) 30 pins
 - (D) 18 pins
 - (E) 10 pins
 - (F) 24 pins
 - (G) 15 pins

4. Twice a number, increased by 5, gives the same result as the number increased by 19. Find the number. [\[Simple Equations\]](#)
 - (A) -14
 - (B) 9
 - (C) 12
 - (D) 14
 - (E) 19
 - (F) 7
 - (G) 24

5. A patient's fever of 39.8 degC falls at a steady rate to the normal 37.0 degC in 4 hours. What is the temperature exactly 3 hours after the fall begins? [\[Heat\]](#)
- (A) 38.0 degC
 - (B) 37.5 degC
 - (C) 39.1 degC
 - (D) 37.0 degC
 - (E) 38.05 degC
 - (F) 37.7 degC
6. In a class the ratio of students who passed to those who failed is 7 : 3. If 18 students failed, how many students are there in all? [\[Comparing Quantities\]](#)
- (A) 54
 - (B) 21
 - (C) 42
 - (D) 30
 - (E) 90
 - (F) 60
7. Three appliances on one socket draw 2 A, 3 A and 2.5 A at once. A 5 A fuse is fitted. What is the total current, and what happens? [\[Electric Current & its Effects\]](#)
- (A) 7.5 A; the bulbs glow brighter
 - (B) 7.5 A; the fuse melts and breaks the circuit
 - (C) 12.5 A; the fuse is safe
 - (D) 2.5 A; the fuse melts
 - (E) 7.5 A; the fuse is safe
 - (F) 3 A; the fuse is safe
 - (G) 5 A; the fuse is safe
8. Three times a number, decreased by 7, equals 26. What is the number? [\[Simple Equations\]](#)
- (A) 33
 - (B) 8
 - (C) 9
 - (D) 19
 - (E) 6.33
 - (F) 13
 - (G) 11
9. Two identical pans of water both start at 30 degC in the sun. The black pan warms by 18 degC and the shiny pan by 6 degC in the same time. How many degrees hotter is the black pan's water now? [\[Heat\]](#)
- (A) 24 degC
 - (B) 3 degC
 - (C) 6 degC
 - (D) 12 degC
 - (E) 18 degC
 - (F) 48 degC

10. Find the simple interest on Rs 4500 at 8% per annum for 2.5 years. [\[Comparing Quantities\]](#)

- (A) Rs 900
- (B) Rs 1125
- (C) Rs 5400
- (D) Rs 360
- (E) Rs 90
- (F) Rs 720
- (G) Rs 1800

11. With one cell a bulb glows at a brightness of 4 units. If three such identical cells are joined in series and the brightness simply adds, what is the brightness? [\[Electric Current & its Effects\]](#)

- (A) 7 units
- (B) 4 units
- (C) 12 units
- (D) 1.33 units
- (E) 6 units
- (F) 8 units
- (G) 16 units

12. One-third of a number, increased by 4, equals 10. What is the number? [\[Simple Equations\]](#)

- (A) 42
- (B) 20
- (C) 2
- (D) 14
- (E) 18
- (F) 30

13. By day, land heats faster than sea, so air over land rises and cooler air flows in from the sea. If this daytime wind blows at 15 km/h, how far does it travel in 12 minutes, and what is the wind called? [\[Heat\]](#)

- (A) 0.8 km; sea breeze
- (B) 1.25 km; sea breeze
- (C) 3 km; monsoon
- (D) 3 km; sea breeze
- (E) 180 km; sea breeze
- (F) 3 km; land breeze

14. Calculate 15% of 40% of 5000. [\[Comparing Quantities\]](#)

- (A) 300
- (B) 275
- (C) 330
- (D) 3000
- (E) 450
- (F) 750
- (G) 2000

15. Which device works mainly on the HEATING effect of electric current, and what is its hot part called?

[Electric Current & its Effects]

- (A) an electric iron; its heating element (coil)
- (B) an electric fan; its element
- (C) an electric bell; its filament
- (D) a fan; its electromagnet
- (E) an electric bell; its coil
- (F) a torch bulb; its magnet

16. A father is 4 times as old as his son. In 5 years he will be only 3 times as old. What is the son's present age? [Simple Equations]

- (A) 8 years
- (B) 40 years
- (C) 5 years
- (D) 12 years
- (E) 20 years
- (F) 15 years
- (G) 10 years

17. Both a steel spoon and a wooden spoon are left in hot soup. The steel spoon soon feels much hotter to touch. Steel is acting as a good _____, while wood is used for handles because it is a good _____. [Heat]

- (A) insulator; conductor
- (B) radiator; reflector
- (C) insulator; insulator
- (D) conductor; conductor
- (E) reflector; absorber
- (F) conductor; insulator

18. A bag is marked Rs 1500. A 20% discount is given, and then 5% tax is added on the discounted price. What does the customer pay? [Comparing Quantities]

- (A) Rs 1185
- (B) Rs 1500
- (C) Rs 1320
- (D) Rs 1200
- (E) Rs 1275
- (F) Rs 1140
- (G) Rs 1260

19. A compass needle placed near a wire deflects (turns) only when the switch is closed and current flows. This shows that an electric current has a: [Electric Current & its Effects]

- (A) cooling effect
- (B) sound effect
- (C) no effect
- (D) lighting effect
- (E) magnetic effect
- (F) chemical effect
- (G) heating effect

20. The sum of three consecutive whole numbers is 72. What is the LARGEST of them? [\[Simple Equations\]](#)

- (A) 72
- (B) 24
- (C) 26
- (D) 23
- (E) 21
- (F) 25
- (G) 48

21. A town's noon temperatures over five days were 31, 34, 29, 33 and 38 degC. What was the mean (average) noon temperature? [\[Heat\]](#)

- (A) 32 degC
- (B) 35 degC
- (C) 33.5 degC
- (D) 165 degC
- (E) 34 degC
- (F) 30 degC
- (G) 33 degC

22. A shopkeeper buys a toy for Rs 250 and sells it for Rs 300. What is the profit percent? [\[Comparing Quantities\]](#)

- (A) 15%
- (B) 16.67%
- (C) 50%
- (D) 25%
- (E) 20%
- (F) 18%

23. An electric bell rings continuously because the current makes an electromagnet that: [\[Electric Current & its Effects\]](#)

- (A) pulls an iron strip to break the circuit, then releases it, again and again
- (B) heats a wire until it glows
- (C) reflects light off a mirror
- (D) stores chemical energy in a cell
- (E) cools the hammer each time
- (F) turns a fan blade round and round

24. Solve: $5(x - 3) = 2x + 6$. [\[Simple Equations\]](#)

- (A) 3
- (B) 5
- (C) 9
- (D) 11
- (E) 7
- (F) -3
- (G) 21

25. Water in a pan is heated only at the bottom; hot water rises and cooler water sinks, forming a current. This way of transferring heat is called _____. If heating 1 L takes 3 minutes at this rate, how long for 2.5 L?

[Heat]

- (A) convection; 7.5 minutes
- (B) convection; 0.83 minutes
- (C) convection; 2.5 minutes
- (D) conduction; 3 minutes
- (E) radiation; 7.5 minutes
- (F) conduction; 7.5 minutes
- (G) convection; 4.5 minutes

26. Express 37.5% as a fraction in its lowest terms. [Comparing Quantities]

- (A) 37/100
- (B) 3/7
- (C) 375/1000
- (D) 5/8
- (E) 3/8
- (F) 1/3
- (G) 15/40

27. In a circuit diagram, a long thin line beside a short thick line stands for which single component? [Electric

Current & its Effects]

- (A) a single cell
- (B) a plain wire
- (C) an open key
- (D) a switch
- (E) a fuse
- (F) an electric bulb

28. Solve: $(2x + 1) / 3 = 5$. [Simple Equations]

- (A) 6
- (B) 7
- (C) 7.5
- (D) 16
- (E) 15
- (F) 8
- (G) 2

29. A freezer cools a packet steadily from 25 degC to -5 degC in 20 minutes. By how many degrees does it cool each minute? [Heat]

- (A) 1.0 degC/min
- (B) 30 degC/min
- (C) 0.75 degC/min
- (D) 1.25 degC/min
- (E) 2.0 degC/min
- (F) 15 degC/min
- (G) 1.5 degC/min

30. Rs 6300 is shared among A, B and C in the ratio 2 : 3 : 4. How much does C get? [\[Comparing Quantities\]](#)

- (A) Rs 1800
- (B) Rs 2800
- (C) Rs 2100
- (D) Rs 1400
- (E) Rs 2400
- (F) Rs 700
- (G) Rs 3600

31. An appliance normally draws a current of 4.5 A. Which fuse rating is the safest sensible choice? [\[Electric Current & its Effects\]](#)

- (A) 1 A
- (B) 5 A
- (C) 4 A
- (D) 10 A
- (E) 3 A
- (F) 0.5 A
- (G) 2 A

32. A number is multiplied by 4 and then 9 is subtracted; the result is 27. What is the number? [\[Simple Equations\]](#)

- (A) 27
- (B) 9
- (C) 4.5
- (D) 18
- (E) 36
- (F) 6

33. We feel the Sun's warmth across empty space where there is no air. The Sun's heat reaches us mainly by ____, because this is the only transfer that needs no medium. [\[Heat\]](#)

- (A) conduction and convection
- (B) radiation
- (C) evaporation
- (D) reflection
- (E) insulation
- (F) conduction
- (G) convection

34. A plant grew from 40 cm to 50 cm in a month. What is the percent increase in its height? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 80%
- (C) 10%
- (D) 12.5%
- (E) 30%
- (F) 2.5%

35. Two cells light a bulb brightly only when they are joined: [\[Electric Current & its Effects\]](#)

- (A) with the bulb between the + ends only
- (B) + to + and - to - (opposing)
- (C) in series facing opposite ways
- (D) with both + ends free
- (E) in series, the + of one to the - of the next (same direction)
- (F) - of one to - of the next
- (G) in a tangle, terminals touching

36. One-third of a number is 8 less than one-half of the same number. Find the number. [\[Simple Equations\]](#)

- (A) 8
- (B) 96
- (C) 12
- (D) 40
- (E) 16
- (F) 24
- (G) 48

37. At night land cools faster than the sea, so wind blows from land to sea; fishing boats ride it out to sea. If this wind blows at 8 km/h for 45 minutes, how far does a boat drift, and what is the wind? [\[Heat\]](#)

- (A) 2.7 km; land breeze
- (B) 8 km; land breeze
- (C) 6 km; cyclone
- (D) 6 km; land breeze
- (E) 360 km; land breeze
- (F) 6 km; sea breeze
- (G) 0.18 km; land breeze

38. A jacket marked Rs 2000 is sold with two successive discounts of 10% and then 20%. What is the final price? [\[Comparing Quantities\]](#)

- (A) Rs 1400
- (B) Rs 1380
- (C) Rs 1620
- (D) Rs 1500
- (E) Rs 1440
- (F) Rs 1600
- (G) Rs 1480

39. A heater's coil raises water's temperature by a steady 3 degC each minute. Starting at 22 degC, how long to reach 49 degC? [\[Electric Current & its Effects\]](#)

- (A) 27 minutes
- (B) 12 minutes
- (C) 18 minutes
- (D) 16.3 minutes
- (E) 6 minutes
- (F) 13 minutes
- (G) 9 minutes

40. A rectangle's length is 3 cm more than its width, and its perimeter is 26 cm. What is the width? [\[Simple](#)

[Equations\]](#)

- (A) 6.5 cm
- (B) 5 cm
- (C) 8 cm
- (D) 4 cm
- (E) 10 cm
- (F) 13 cm

41. In summer we prefer light-coloured, loose clothes. The best pair of reasons is: [\[Heat\]](#)

- (A) dark colours absorb less heat; tight clothes trap sweat
- (B) dark colours reflect heat; loose clothes trap heat
- (C) loose clothes conduct heat to us; light colours store heat
- (D) light colours reflect cold; loose clothes block air
- (E) light colours absorb more heat; loose clothes warm us
- (F) light colours absorb less heat; loose clothes let air move and cool us
- (G) light colours make heat; tight clothes are cooler

42. A sum of Rs 5000 grows to Rs 5900 in 2 years under simple interest. What is the rate per annum?

[\[Comparing Quantities\]](#)

- (A) 10%
- (B) 11.8%
- (C) 8%
- (D) 45%
- (E) 18%
- (F) 4.5%
- (G) 9%

43. A bulb glows only dimly with one cell. The simplest way to make it glow brighter is to: [\[Electric Current & its](#)

[Effects\]](#)

- (A) remove the cell
- (B) add more cells in series
- (C) use a longer connecting wire
- (D) open the switch a little
- (E) use a thicker bulb glass
- (F) wind the wire into a coil
- (G) paint the bulb black

44. Reena is 6 years older than her brother. In 4 years the sum of their ages will be 30. What is Reena's present age? [\[Simple Equations\]](#)

- (A) 10 years
- (B) 8 years
- (C) 14 years
- (D) 16 years
- (E) 20 years
- (F) 11 years
- (G) 12 years

45. Equal cups of water, one at 20 degC and one at 60 degC, are mixed together. The temperature of the mixture is about: [\[Heat\]](#)
- (A) 80 degC
 - (B) 45 degC
 - (C) 50 degC
 - (D) 30 degC
 - (E) 40 degC
 - (F) 20 degC
46. A 2.5 L bottle is 60% full. How many millilitres of liquid does it hold right now? [\[Comparing Quantities\]](#)
- (A) 1500 mL
 - (B) 1600 mL
 - (C) 150 mL
 - (D) 1000 mL
 - (E) 2500 mL
 - (F) 600 mL
 - (G) 1250 mL
47. An electromagnet is preferred over an ordinary bar magnet in a scrap-yard crane mainly because: [\[Electric Current & its Effects\]](#)
- (A) it lifts without any current
 - (B) it attracts wood and plastic too
 - (C) it works only when switched off
 - (D) it never needs any electricity
 - (E) it can be switched on to lift and off to drop the load
 - (F) it is always weaker than a bar magnet
 - (G) it cannot ever lose its magnetism
48. Pencils cost Rs 4 each and erasers Rs 6 each. Riya buys 5 pencils and some erasers for a total of Rs 38. How many erasers did she buy? [\[Simple Equations\]](#)
- (A) 3
 - (B) 1
 - (C) 2
 - (D) 6
 - (E) 5
 - (F) 9
 - (G) 4
49. Telephone wires between poles are deliberately left a little loose (sagging). The main reason is: [\[Heat\]](#)
- (A) sagging wires keep birds away
 - (B) in summer they contract and pull the poles down
 - (C) in winter they contract and would snap if strung tight
 - (D) tight wires expand and break in winter
 - (E) loose wires carry current faster
 - (F) sagging stops the wires rusting
 - (G) loose wires conduct less heat

50. The ratio of boys to girls in a club is 5 : 4. There are 12 more boys than girls. How many girls are there?

[Comparing Quantities]

- (A) 36
- (B) 48
- (C) 60
- (D) 54
- (E) 12
- (F) 24
- (G) 96

51. A fuse wire melts and breaks the circuit when too large a current flows. Its main purpose is to: [Electric

Current & its Effects]

- (A) measure the temperature of the room
- (B) store energy for later use
- (C) protect the wiring and appliances from overheating and fire
- (D) increase the current in the circuit
- (E) make the bulbs glow brighter
- (F) cool the wires down by melting
- (G) produce a magnetic field on purpose

52. The sum of two numbers is 25, and the larger is 4 times the smaller. What is the SMALLER number?

[Simple Equations]

- (A) 20
- (B) 21
- (C) 9
- (D) 10
- (E) 4
- (F) 5
- (G) 6.25

53. On a cold morning a knitted woollen sweater feels much warmer than a thin cotton shirt. The main reason is that the wool: [Heat]

- (A) trapped air conducts heat to us quickly
- (B) wool absorbs cold from the body
- (C) wool reflects all the cold away
- (D) wool is a very good conductor of heat
- (E) wool stops air from touching the skin entirely
- (F) they trap air, and trapped air is a poor conductor of heat
- (G) wool itself produces heat

54. A student's test score improved from 60% to 75%. By what percent did the score increase? [Comparing

Quantities]

- (A) 25%
- (B) 75%
- (C) 20%
- (D) 12.5%
- (E) 40%
- (F) 18%

- 55.** A toy needs the push of 4 cells joined in series. Each cell provides 1.5 'units' of push. What is the total push? [\[Electric Current & its Effects\]](#)
- (A) 1.5 units
 - (B) 5.5 units
 - (C) 6 units
 - (D) 4 units
 - (E) 0.375 units
 - (F) 2.5 units
 - (G) 7.5 units
- 56.** Solve: $(x - 4) / 2 + 3 = 7$. [\[Simple Equations\]](#)
- (A) 14
 - (B) 6
 - (C) 12
 - (D) 10
 - (E) 20
 - (F) 16
 - (G) 8
- 57.** Which is colder, -12 degC or -7 degC, and by how many degrees? [\[Heat\]](#)
- (A) both are equally cold
 - (B) -7 degC, by 5 degC
 - (C) -12 degC, by 7 degC
 - (D) -7 degC, by 19 degC
 - (E) -12 degC, by 5 degC
 - (F) -12 degC, by 19 degC
 - (G) -12 degC, by 12 degC
- 58.** A shopkeeper buys 12 identical pens for Rs 96 and sells them all at Rs 10 each. What is the total profit? [\[Comparing Quantities\]](#)
- (A) Rs 12
 - (B) Rs 96
 - (C) Rs 2
 - (D) Rs 120
 - (E) Rs 8
 - (F) Rs 20
 - (G) Rs 24
- 59.** A bulb fails to light and you find the switch is open. 'An open switch' means that: [\[Electric Current & its Effects\]](#)
- (A) the bulb is broken inside
 - (B) the wires are too thick
 - (C) current flows but the bulb is off
 - (D) the cell has run out completely
 - (E) the circuit is complete, so current flows
 - (F) the circuit is broken, so no current can flow
 - (G) the fuse has melted

60. Seven added to twice a number gives 33. What is the number? [\[Simple Equations\]](#)

- (A) 13
- (B) 6.5
- (C) 16.5
- (D) 20
- (E) 23
- (F) 40
- (G) 26

61. A liquid is at 40 degC. Its temperature is then increased by 25%. The new temperature is: [\[Heat\]](#)

- (A) 65 degC
- (B) 50 degC
- (C) 45 degC
- (D) 10 degC
- (E) 25 degC
- (F) 55 degC

62. An article is sold for Rs 660, giving a 10% profit. What was its cost price? [\[Comparing Quantities\]](#)

- (A) Rs 726
- (B) Rs 726.0
- (C) Rs 600.0
- (D) Rs 600
- (E) Rs 594
- (F) Rs 666
- (G) Rs 650

63. A bulb gives light because the current heats its very thin wire (filament) until it: [\[Electric Current & its Effects\]](#)

- (A) melts and breaks at once
- (B) glows white-hot and gives off light
- (C) turns into a magnet
- (D) cools the surroundings
- (E) stays completely cold
- (F) stores the current as charge

64. The sum of three consecutive EVEN numbers is 90. What is the MIDDLE number? [\[Simple Equations\]](#)

- (A) 32
- (B) 28
- (C) 29
- (D) 45
- (E) 20
- (F) 60
- (G) 30

- 65.** A laboratory thermometer is marked from -10°C to 110°C . How many degrees does its full scale span? [\[Heat\]](#)
- (A) 130°C
 - (B) 120°C
 - (C) 90°C
 - (D) 100°C
 - (E) 120.5°C
 - (F) 10°C
- 66.** A salt solution mixes salt and water in the ratio 1 : 24 by mass. How much salt is in 500 g of the solution? [\[Comparing Quantities\]](#)
- (A) 480 g
 - (B) 30 g
 - (C) 25 g
 - (D) 100 g
 - (E) 50 g
 - (F) 24 g
 - (G) 20 g
- 67.** For the same current, a long thin wire gets hotter than a short thick wire because the long thin wire has: [\[Electric Current & its Effects\]](#)
- (A) more resistance to the current
 - (B) less resistance to the current
 - (C) a stronger magnet
 - (D) more switches
 - (E) more cells inside it
 - (F) no current at all
 - (G) a built-in fuse
- 68.** If 8 is subtracted from 3 times a number, the result equals the number plus 12. Find the number. [\[Simple Equations\]](#)
- (A) 5
 - (B) 10
 - (C) 2
 - (D) 4
 - (E) 6
 - (F) 20
 - (G) -10
- 69.** Heat is sent along a copper rod and an identical glass rod from one end. Heat reaches the far end of copper in 8 s, and copper carries heat about 4 times faster than glass. About how long for glass? [\[Heat\]](#)
- (A) 4 s
 - (B) 24 s
 - (C) 12 s
 - (D) 16 s
 - (E) 2 s
 - (F) 8 s
 - (G) 32 s

70. After a 25% discount, a school bag costs Rs 900. What was its marked price? [\[Comparing Quantities\]](#)

- (A) Rs 1150
- (B) Rs 1000
- (C) Rs 675
- (D) Rs 1200
- (E) Rs 1125
- (F) Rs 1500
- (G) Rs 1350

71. An electromagnet lifts 21 nails using 70 turns of wire. To lift just 9 nails (lift proportional to turns), about how many turns are needed? [\[Electric Current & its Effects\]](#)

- (A) 90 turns
- (B) 58 turns
- (C) 30 turns
- (D) 27 turns
- (E) 63 turns
- (F) 45 turns

72. A number decreased by its one-fourth equals 18. What is the number? [\[Simple Equations\]](#)

- (A) 24
- (B) 22.5
- (C) 72
- (D) 18
- (E) 4.5
- (F) 14.4
- (G) 20

73. Over 4 days the mean noon temperature was 35 degC. Three of the days read 33, 36 and 34 degC. What was the fourth day's temperature? [\[Heat\]](#)

- (A) 38 degC
- (B) 35 degC
- (C) 33 degC
- (D) 103 degC
- (E) 36 degC
- (F) 37 degC
- (G) 34 degC

74. A town's population fell from 8000 to 7000 in a year. What is the percent decrease? [\[Comparing Quantities\]](#)

- (A) 87.5%
- (B) 14.28%
- (C) 10%
- (D) 15%
- (E) 1000%
- (F) 12.5%

75. Three heaters, each drawing 6 A, are plugged into one socket. The total current is found, and the line has a 10 A fuse. What happens? [\[Electric Current & its Effects\]](#)

- (A) the total is 10 A, so nothing happens
- (B) the total is 18 A, so the fuse melts and cuts the supply
- (C) the total is 18 A, so the fuse is safe
- (D) the total is 2 A, so the fuse melts
- (E) the heaters share 6 A safely
- (F) the total is 6 A, so the fuse is safe
- (G) the fuse makes the heaters hotter

76. The cost of 3 notebooks and 2 pens is Rs 70. A notebook costs Rs 16. What does one pen cost? [\[Simple Equations\]](#)

- (A) Rs 11
- (B) Rs 22
- (C) Rs 5
- (D) Rs 13
- (E) Rs 27
- (F) Rs 9
- (G) Rs 8

77. On a hot afternoon at the beach, which is warmer and which way does the breeze blow? [\[Heat\]](#)

- (A) the sea is warmer; the breeze blows from sea to land
- (B) the sand is warmer; the breeze blows from land to sea
- (C) the sea is warmer; the breeze blows in circles
- (D) the sand is warmer; the breeze blows from sea to land
- (E) the sand is warmer; the breeze blows straight up
- (F) the sea is warmer; the breeze blows from land to sea

78. Find the simple interest on Rs 1200 at 10% per annum for 9 months. [\[Comparing Quantities\]](#)

- (A) Rs 9
- (B) Rs 135
- (C) Rs 120
- (D) Rs 100
- (E) Rs 108
- (F) Rs 1080
- (G) Rs 90

79. You should never touch an electric switch with wet hands because: [\[Electric Current & its Effects\]](#)

- (A) water makes the bulb brighter
- (B) water turns the current off
- (C) water conducts electricity and you could get a dangerous shock
- (D) water makes the switch rust instantly
- (E) wet hands are too cold for the switch
- (F) wet hands cannot press a switch
- (G) water stops the magnetic effect

- 80.** A man is 3 times as old as his son. Twelve years ago he was 7 times as old. What is the son's present age? [\[Simple Equations\]](#)
- (A) 54 years
 - (B) 12 years
 - (C) 9 years
 - (D) 18 years
 - (E) 36 years
 - (F) 6 years
- 81.** A black box and a white box sit in the sun. The black box warms from 30 degC to 54 degC; the white box rises only half as much. What is the white box's final temperature? [\[Heat\]](#)
- (A) 24 degC
 - (B) 27 degC
 - (C) 12 degC
 - (D) 54 degC
 - (E) 42 degC
 - (F) 36 degC
- 82.** In an exam, 30% failed in maths, 20% failed in science, and 10% failed in BOTH. What percent of students passed in both subjects? [\[Comparing Quantities\]](#)
- (A) 40%
 - (B) 70%
 - (C) 30%
 - (D) 10%
 - (E) 50%
 - (F) 60%
 - (G) 80%
- 83.** The energy that drives the current in a working torch comes from: [\[Electric Current & its Effects\]](#)
- (A) energy made by the bulb's glow
 - (B) magnetic energy in the wires
 - (C) chemical energy stored in the cells
 - (D) motion energy of the user
 - (E) sound energy from the click
 - (F) light energy from the bulb
 - (G) heat energy from the switch
- 84.** Twice the sum of a number and 3 equals 22. What is the number? [\[Simple Equations\]](#)
- (A) 11
 - (B) 9.5
 - (C) 5
 - (D) 6
 - (E) 16
 - (F) 8

85. At a hill station it was 4 degC at 6 pm and the temperature dropped a steady 2 degC every hour. At what time did it reach -6 degC? [\[Heat\]](#)

- (A) 9 pm
- (B) 5 pm
- (C) midnight
- (D) 11 pm
- (E) 12 noon
- (F) 1 am

86. A recipe for 4 people uses 300 g of flour. Keeping the same ratio, how much flour is needed for 7 people? [\[Comparing Quantities\]](#)

- (A) 2100 g
- (B) 450 g
- (C) 600 g
- (D) 75 g
- (E) 525 g
- (F) 700 g

87. Winding more turns of wire around an iron nail (same current) makes the electromagnet: [\[Electric Current & its Effects\]](#)

- (A) lose all magnetism
- (B) change into a fuse
- (C) switch off
- (D) cooler only
- (E) weaker
- (F) stronger
- (G) reverse the current

88. When a number is divided by 5 and 3 is added, the result is 10. What is the number? [\[Simple Equations\]](#)

- (A) 35
- (B) 15
- (C) 7
- (D) 13
- (E) 50
- (F) 26
- (G) 65

89. Which set contains ONLY good conductors of heat? [\[Heat\]](#)

- (A) copper, wool, glass
- (B) iron, air, rubber
- (C) iron, copper, paper
- (D) aluminium, water, cork
- (E) plastic, glass, wood
- (F) iron, copper, aluminium

90. A student scored 380 out of 500. What percent is this, and is it above the 80% distinction mark?

[Comparing Quantities]

- (A) 80%; below distinction
- (B) 82%; above distinction
- (C) 75.6%; below distinction
- (D) 76%; above distinction
- (E) 74%; below distinction
- (F) 76%; below distinction
- (G) 38%; below distinction

91. Among a geyser, a ceiling fan and a doorbell, which one chiefly uses the HEATING effect of current?

[Electric Current & its Effects]

- (A) the ceiling fan
- (B) none of them
- (C) the doorbell only
- (D) all three equally
- (E) the doorbell
- (F) the geyser

92. A number decreased by its two-fifths equals 18. What is the number? [Simple Equations]

- (A) 25.7
- (B) 30
- (C) 7.2
- (D) 20
- (E) 36
- (F) 12

93. A thermos (vacuum) flask keeps a drink hot for hours. The vacuum (empty gap) between its double walls mainly stops which kinds of heat transfer? [Heat]

- (A) conduction only
- (B) radiation only
- (C) conduction and convection
- (D) convection only
- (E) radiation and convection
- (F) none of them
- (G) all three completely

94. A salary of Rs 20000 is raised by 10%, and then by a further 5% on the new amount. What is the final salary? [Comparing Quantities]

- (A) Rs 21500
- (B) Rs 22500
- (C) Rs 23100
- (D) Rs 24000
- (E) Rs 23000
- (F) Rs 25000

- 95.** A line is protected by a 6 A fuse. Appliances drawing 2 A and 3 A run fine. A third appliance drawing 2 A is then switched on. What happens? [\[Electric Current & its Effects\]](#)
- (A) the total becomes 7 A, so the fuse melts
 - (B) the total becomes 4 A, so the fuse melts
 - (C) the total becomes 7 A, so it is safe
 - (D) the fuse makes them all brighter
 - (E) the total becomes 7 A, so nothing changes
 - (F) the total stays 5 A, so it is safe
- 96.** Six times a number equals the number increased by 30. What is the number? [\[Simple Equations\]](#)
- (A) 30
 - (B) 5
 - (C) 6.5
 - (D) 6
 - (E) -6
 - (F) 36
 - (G) 7.5
- 97.** A metal bar is 200 cm long. On strong heating it expands by 0.5% of its length. What is its new length? [\[Heat\]](#)
- (A) 205 cm
 - (B) 201 cm
 - (C) 210 cm
 - (D) 200.5 cm
 - (E) 100 cm
 - (F) 201.5 cm
- 98.** If 3 out of every 8 fruits in a crate are rotten, what percent of the fruits are good? [\[Comparing Quantities\]](#)
- (A) 62.5%
 - (B) 37.5%
 - (C) 50%
 - (D) 32.5%
 - (E) 80%
 - (F) 65%
 - (G) 60%
- 99.** For a bulb to glow, the circuit must be: [\[Electric Current & its Effects\]](#)
- (A) broken at one point
 - (B) missing the cell
 - (C) connected at one end only
 - (D) made only of insulators
 - (E) complete (closed), with the cell connected at both ends of the path
 - (F) open at the switch
 - (G) without any wire

100. The sum of three consecutive multiples of 5 is 75. What is the SMALLEST of them? [\[Simple Equations\]](#)

- (A) 30
- (B) 20
- (C) 5
- (D) 15
- (E) 10
- (F) 25